Fall 2026.

Physics 515. Statistical Mechanics.
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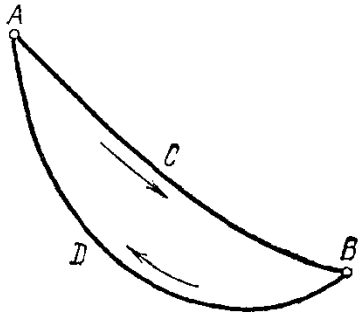
Homework # 1.

Due to October 8th, 2026.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it. Provide me with your code and final graphs when you use computational methods.

1. Show that for any isotropic, homogeneous material $\kappa_S = \frac{C_V}{C_p}\kappa_T$, where $\kappa_S = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_S$ is an adiabatic compressibility, $\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$ is an isothermal compressibility, and C_p and C_V are molar heat capacities under a constant pressure and volume. Comments: do not assume that this is an ideal gas; C_p, and C_V might be heat capacities per mole, mass unit etc.

2. A polytropic process is such a process that a corresponding molar heat capacity C stays constant. Prove that a polytrope equation for an ideal gas is $pV^n = \text{constant}$, where $n = \frac{C-C_p}{C-C_V}$ is a polytropic index. Calculate the index for adiabatic, isothermal, isobaric, and isochoric processes.



3. Prove that a polytrope and isotherm can cross at one point maximum. Hint: assume that A and B are two nearest crossing points; apply the Clausius equality to the thermodynamical cycle in the figure where ADB is the polytrope, and ACB is the isotherm; get a contradiction.

4. Consider an ideal gas of massless particles (photons, for example). For such a gas the internal energy density $u(T)$ (the energy per volume) is connected to the pressure p as following $p = \frac{1}{3}u(T)$.

a) Determine the function $u(T)$. It is nothing but the Stefan-Boltzmann Law.
b) Calculate the entropy density of the gas.
c) Derive the adiabata equation and expression for a constant volume specific heat capacity.
d) Discuss a Carnot cycle with the photons gas as a working material.
e) Prove the Clausius equity using this cycle.

5. Another popular equation of real gases is the Dieterici equation $p(v-b) = RTe^{-a/RTv}$, where v is a molar volume. Follow the pattern of the van der Waals equation analysis we performed in the class.

a) Find criteria for unstable isotherms.
b) Calculate critical parameters of the equation.
c) Write the equation down in the reduced form.
d) Discuss the Joule-Thomson process using the Dieterici equation.
e) Derive the inversion temperature formula.

6. Prove the following thermodynamical relations:

a) $\left(\frac{\partial S}{\partial T}\right)_V = \frac{1}{T} C_V$, $\left(\frac{\partial S}{\partial T}\right)_p = \frac{1}{T} C_p$.

b) $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$, $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$.

c) $C_p - C_V = T\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial p}{\partial T}\right)_V$.

d) $\left(\frac{\partial E}{\partial V}\right)_T = \left(\frac{\partial E}{\partial p}\right)_T \left(\frac{\partial p}{\partial V}\right)_T$.